ZERO-SHOT COORDINATION UNDER PARTIAL OBSERVABILITY BY RECONSTRUCTION UNCERTAINTY

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ABSTRACT

The successful deployment of AI agents into complex physical environments necessitates the ability to robustly coordinate with diverse partners, ranging from other AI agents to humans. Consequently, agents must possess the capability to cooperate with novel partners encountered only at test time—a scenario where conventional Self-Play methods frequently fail due to their tendency to rely on specific, non-generalizable conventions. This challenge has motivated the Zero-Shot Coordination (ZSC) paradigm. While recent population-based approaches have demonstrated significant progress, they often exhibit performance degradation in stochastic environments. This limitation largely arises because agents rely heavily on partial information provided by their partners, rather than actively inferring the underlying global state. Although reconstructing full observations offers a potential solution to bridge this information gap, it introduces a critical risk - agents may become overconfident in incorrect predictions derived from limited data. To overcome this dilemma while ensuring real-time applicability, we propose a novel evidential reconstruction pipeline. This framework enables the agent to explicitly quantify predictive uncertainty in a single forward pass, avoiding the computational burden of multiple inferences. We empirically validate our approach on the OvercookedV2 benchmark, specifically targeting stochastic ZSC settings. Our extensive quantitative and visual results demonstrate that our method significantly improves coordination by allowing agents to adaptively handle uncertainty.

1 INTRODUCTION

Recently, Artificial Intelligence (AI) has achieved remarkable success in areas such as LLMs, image/video generation, agentic AI, and computer vision. However, deploying AI from virtual environments to the physical world remains an open problem, despite significant progress in autonomous driving. One of the primary hurdles is the necessity for agents to interact with novel entities, including humans, other AI agents, and even animals. Agents trained via Self-Play (SP), where a single partner (either the agent itself or another trainable agent) is present during training, often fail to cooperate with novel partners unseen during training (Hu et al., 2020; Stone et al., 2010), notwithstanding the milestone achievements of SP in games like Go (Silver et al., 2016).

To address this, paradigms such as Zero-Shot Coordination (ZSC) (Hu et al., 2020) and ad-hoc teamwork (Stone et al., 2010) have been proposed. Recent approaches focusing on population-based training with either implicit (Strouse et al., 2021; Yu et al., 2023; Sarkar et al., 2023; Cui et al., 2021) or explicit policy diversity (Lupu et al., 2021; Zhao et al., 2023) have demonstrated the ability to coordinate with unseen AI agents, behavior-cloned human-proxy models (Carroll et al., 2019), and even real humans (Wang et al., 2024). However, Gessler et al. (2025) argue that existing methods struggle when stochasticity is introduced, such as random recipes and partial observability. In such settings, an agent must identify and focus on the underlying context (e.g., the recipe) in accordance with its partner's actions. Yet, existing methods rely on inferring Theory of Mind (ToM) (Premack & Woodruff, 1978; Rabinowitz et al., 2018) either implicitly via population-based methods or explicitly through partner modeling (Ribeiro et al., 2023; Barrett & Stone, 2015; Chen et al., 2020; Gu et al., 2022). This limitation makes it difficult to focus on critical interactions or states under partial observability.

In contrast, humans often act by inferring key environmental factors—specifically the recipe with own confidence(Barthelmé & Mamassian, 2010; Somatori & Kunisato, 2022) in our setup. This raises the question: what if we reconstruct the full observation from a trajectory of partial observations? However, an agent might act on erroneous predictions due to the impossibility of perfect reconstruction. Therefore, it is crucial to measure the uncertainty of these predictions. While Bayesian Neural Networks (BNNs) (MacKay, 1992; Neal, 1995; Gal & Ghahramani, 2016; Blundell et al., 2015) are a prominent approach for uncertainty estimation, they require multiple inferences to measure epistemic uncertainty. This computational cost makes them unsuitable for AI agents in the physical world requiring real-time interaction. Deep evidential learning methods (Sensoy et al., 2018; Amini et al., 2020) offer a more suitable alternative for uncertainty-aware prediction.

Thus, we propose Evidential Reconstruction for Zero-Shot Coordination. Since our method is designed to be a modular extension of existing approaches, we build our algorithm upon Fictitious Co-Play (FCP) (Strouse et al., 2021), a standard baseline for ZSC.

2 Approach

2.1 Problem Formulation

We formulate our problem within the framework of Decentralized Partially Observable Markov Decision Processes (Dec-POMDPs) Bernstein et al. (2000), extended to the zero-shot coordination setting. A Dec-POMDP is defined by a tuple $\mathcal{G} = \langle \mathcal{N}, \mathcal{S}, \mathcal{A}, \mathcal{T}, \Omega, \mathcal{O}, \mathcal{R}, \gamma \rangle$.

Environment and Dynamics. Let $\mathcal{N} = \{1, \dots, N\}$ denote the set of N agents. At each time step t, the global state is $s_t \in \mathcal{S}$. The joint action space is $\mathcal{A} = \times_{i=1}^{N} \mathcal{A}^i$, where $\mathcal{A}^i$ is the discrete action space for agent i. The environment transitions to the next state s_{t+1} according to the transition function $\mathcal{T} : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})$, denoted as $P(s_{t+1}|s_t, \boldsymbol{a}_t)$, where $\boldsymbol{a}_t = (a_t^1, \dots, a_t^N)$ represents the joint action.

Observations. Due to partial observability, agent i cannot access the global state s_t directly. Instead, it receives a local observation o_t^i determined by the observation function $\mathcal{O} : \mathcal{S} \times \mathcal{A} \to \Delta(\Omega^i)$. We define the observation as a vectorized input $\mathbf{o}_t^i \in \mathbb{R}^d$, which provides partial information about the environment. To mitigate the uncertainty arising from partial observability, each agent maintains an action-observation history $\tau_t^i = (\mathbf{o}_0^i, a_0^i, \dots, a_{t-1}^i, \mathbf{o}_t^i) \in \mathcal{H}^i$.

Policy and Objective. Each agent i acts according to a policy $\pi^i : \mathcal{H}^i \to \Delta(\mathcal{A}^i)$. We denote the set of all partners interacting with agent i as $-i$, and their joint policy as π^{-i}. The agents share a common reward function $r(s_t, \boldsymbol{a}_t)$, and the goal is to maximize the discounted cumulative return $R_t = \sum_{k=0}^{T-t} \gamma^k r_{t+k}$, where γ is the discount factor.

In the population-based training regime, the agent must robustly cooperate with diverse partners π^{-i} drawn from a distribution $\mathcal{D}$, which were frozen during training. Thus, the objective of agent i is to maximize the expected return over this distribution:

$$\pi^{i,*} = \arg\max_{\pi^i} \mathbb{E}_{\pi^{-i} \sim \mathcal{D}} \left[\mathbb{E}_{\tau \sim (\pi^i, \pi^{-i})} \left[\sum_{t=0}^{T} \gamma^t r(s_t, \boldsymbol{a}_t) \right] \right] \tag{1}$$

2.2 Fictitious Co-Play

Fictitious Co-Play (FCP) (Strouse et al., 2021) is a foundational algorithm in cooperative AI and serves as a standard baseline for Zero-Shot Coordination (ZSC). The structure of FCP is simple yet effective. It builds a diverse population of partner agents by training them with different random seeds. In cooperative benchmarks like Overcooked, there are often multiple potential conventions (local optima). Agents trained with different seeds naturally converge to different conventions, thereby creating diversity.

Furthermore, a robust agent must coordinate not only with experts but also with partners of varying skill levels, such as sub-optimal AI agents or humans. To address this, FCP constructs its partner pool using checkpoints from various stages of training (e.g., initial, mid-training, and fully converged models). Finally, a best-response agent is trained against this frozen pool of diverse partners. This partner pool would be a $\mathcal{D}$ in the 1. We build our method upon FCP because it is the standard paradigm in the field, and following researches only add explicit diversity to make partner. Also, our proposed method is modular, making it compatible with other algorithms as well.

2.3 Evidential Reconstruction with Uncertainty

The objective of the reconstruction module is to infer the full global state information from the trajectory of local observations. Since the environment representation consists of vectorized integers, we formulate the reconstruction as a sequence modeling and classification task. We employ a decoder-only Transformer (Vaswani et al., 2017; Brown et al., 2020) as the backbone due to its robust performance in processing sequential data and its training efficiency.

Instead of standard deterministic classification, we utilize Deep Evidential Classification (Sensoy et al., 2018) to estimate predictive uncertainty. This is implemented by replacing the final softmax layer with an output layer that predicts the parameters $\boldsymbol{\alpha}$ of a Dirichlet distribution $D(\mathbf{p}|\boldsymbol{\alpha})$. The loss function is formulated as:

$$\mathcal{L}(\theta) = \mathcal{L}_{MSE}(\theta) + \lambda \mathrm{KL}\left[D(\mathbf{p}|\tilde{\boldsymbol{\alpha}}) \parallel D(\mathbf{p}|\mathbf{1})\right], \tag{2}$$

where $\tilde{\boldsymbol{\alpha}} = \mathbf{y} + (1-\mathbf{y}) \odot \boldsymbol{\alpha}$ is the adjusted parameter vector for the KL regularization, and $\mathbf{y}$ is the ground-truth label. The MSE component decomposes into the squared error of the expected probability and the variance:

$$\begin{aligned} \mathcal{L}_{MSE}(\theta) &= \sum_{k=1}^{K} \left((y_k - \mathbb{E}[p_k])^2 + \mathrm{Var}(p_k) \right) && (3) \\ &= \sum_{k=1}^{K} \left((y_k - \hat{p}_k)^2 + \frac{\hat{p}_k(1-\hat{p}_k)}{S+1} \right). && (4) \end{aligned}$$

Here, $S = \sum_{k=1}^{K} \alpha_k$ denotes the total strength of the Dirichlet distribution (total evidence), and $\hat{p}_k = \alpha_k / S$ represents the expected probability.

Finally, the reconstructed state is determined by $\mathrm{argmax}(\boldsymbol{\alpha})$. We can explicitly derive the *epistemic uncertainty* (refelcting confident) and the *aleatoric uncertainty* (variance) for each grid cell as follows:

$$\text{Epistemic Uncertainty:} \quad \mathbf{u} = \frac{K}{S} \tag{5}$$

$$\text{Aleatoric Uncertainty:} \quad \mathbf{a} = \frac{\hat{\mathbf{p}}(1-\hat{\mathbf{p}})}{S+1} \tag{6}$$

We pre-train reconstruction model with trajectory collected with SP of partners as following pre-training fine-tuning regime(Devlin et al., 2019).

2.4 Training the Cooperative Agent

We train our cooperative policy by integrating the pre-trained evidential reconstruction module into the FCP framework where evidential reconstruction module is also updated. At each timestep, the agent receives a partial observation and simultaneously reconstructs the global state using the reconstruction model. To provide the agent with cues regarding the reliability of the predicted state, we aggregate the estimated epistemic and aleatoric uncertainties into a single channel and concatenate it with the reconstructed state.Recognizing that reconstruction may be unstable or inaccurate in unobserved regions, we employ a dual-branch architecture that utilizes both the raw local observation and the uncertainty-augmented reconstructed state. Features from both inputs are extracted via separate CNN encoders (LeCun & Bengio, 1995) and concatenated. These fused features are then processed by an LSTM layer (Hochreiter & Schmidhuber, 1997) to capture temporal dependencies. Finally, the recurrent output is fed into Actor and Critic heads (Konda & Tsitsiklis, 1999; Sutton et al., 1998) to produce action logits and state values.We optimize the policy using Independent PPO (IPPO) (de Witt et al., 2020; Schulman et al., 2017), treating the problem as a decentralized multi-agent task. The action sampling process is formulated as:

$$a_t \sim \pi_\theta \left(\cdot \mid \mathrm{LSTM}(\phi_{rec}(\hat{s}_t, \mathbf{u}t), \phi loc(o_t)) \right), \tag{7}$$

where ϕ_{rec} and ϕ_{loc} denote the encoders for the reconstructed and local streams, respectively.Crucially, since the reconstructed state and its uncertainty inherently reflect the unobserved behaviors of partner agents, our method effectively performs implicit partner modeling. This distinguishes our approach from existing population-based methods by grounding coordination in explicit state reasoning under uncertainty.

3 Experimental Design

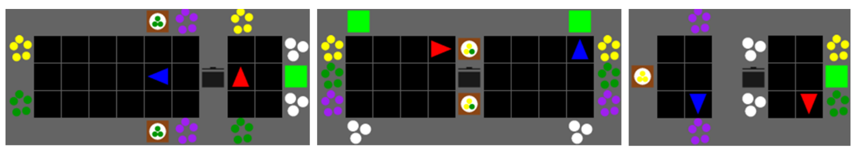


Figure 1: Environment layouts: Demo Cook Simple (Left), Custom Layout (Mid), and Test Time Coord Simple (Right).

Environment. We conduct our experiments on OvercookedV2 (Gessler et al., 2025), an extended version of the collaborative cooking benchmark Overcooked (Carroll et al., 2019). To accelerate training, we utilize the JaxMARL (Rutherford et al., 2024) framework, which enables high-throughput simulation entirely on the GPU via JAX (Bradbury et al., 2018), effectively eliminating CPU-GPU data transfer bottlenecks. The observation space consists of multiple vector channels representing the current status of objects on the grid.In this environment, two agents must coordinate to collect ingredients, cook, and serve dishes. Unlike the original benchmark, OvercookedV2 introduces dynamic recipes and partial observability. We evaluate our method on three distinct layouts shown in Figure 1.

The *Demo Cook Simple* (left) and *Test Time Coord Simple* (right) layouts, proposed by Gessler et al. (2025), focus on asymmetric coordination: one agent observes the recipe and signals it, while the other infers the partner's intention to cook and deliver.Notably, we modify the *Test Time Coord Simple* layout to resample the recipe only at the beginning of each episode, rather than after every delivery. This modification is crucial because real-time adaptation to frequently changing recipes is nearly impossible for strangers, making it a suitable testbed for emergent communication (Bansal et al., 2018). In the *Custom Layout* (mid), the recipe consists of all possible permutations of three ingredients without negative rewards for wrong deliveries. Following the original configuration, the view radius is set to 2, providing a 5×5 field of view centered on the agent.

Partner Pool Creation. We generate a diverse pool of partners using Independent PPO (IPPO) (de Witt et al., 2020; Schulman et al., 2017) with parameter sharing. The agent architecture comprises a CNN encoder (LeCun & Bengio, 1995) followed by an LSTM (Hochreiter & Schmidhuber, 1997) to handle partial observability. We employ a specialized CNN structure: initial layers use a kernel size of 1 to process pixel-wise spatial information, while deeper layers use a kernel size of 3. We exclude checkpoints from the early stages of training, as random policies in this complex environment fail to provide meaningful interaction signals, unlike in the original Overcooked (Carroll et al., 2019).

Implementation Details. To facilitate learning, we employ dense reward shaping for sub-tasks such as placing correct ingredients in the pot and plating cooked dishes. These shaping rewards are annealed over time based on the total step count. We use Generalized Advantage Estimation (GAE) (Schulman et al., 2018) for value estimation. For the reconstruction model, we incorporate Rotary Positional Embeddings (RoPE) (Su et al., 2024) to capture relative positions. To optimize inference latency during rollouts, we utilize KV caching (Vaswani et al., 2017). Furthermore, to improve sample efficiency when fine-tuning the reconstruction module, we employ an experience replay buffer (Mnih et al., 2015) that stores the last K episodes. The AdamW optimizer (Kingma & Ba, 2014; Loshchilov & Hutter, 2017a) with cosine learning rate decay(Loshchilov & Hutter, 2017b) is used throughout the training pipeline.

4 EXPERIMENTAL RESULTS

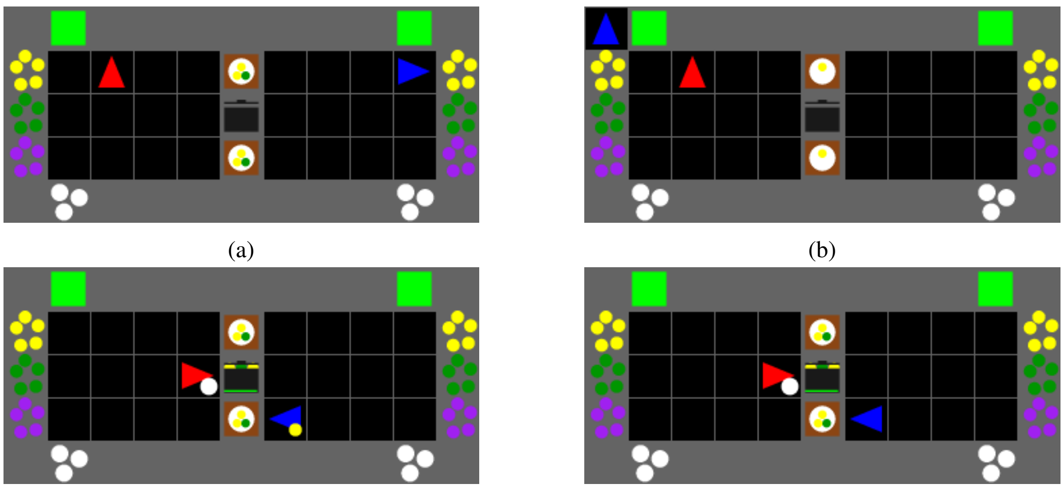

(a) (b) (c) (d)

Figure 2: Reconstruction results on custom layout. (A) The initial frame of real observation. (B) The initial frame of prediction. (C) The frame observing partner agent of real observation. (D) The frame observing partner agent of prediction.

Qualitative results of reconstruction model We first measure qualitative results of reconstruction model. As you can see in the 2, The initial prediction2b where agent haven't observed any recipe or partner agent is incorrect while the structure of map is perfectly predicted by comparing with2a. It is natural and proper way of our assumption Where prediction after observing recipe and partner agent is almost correct2d.

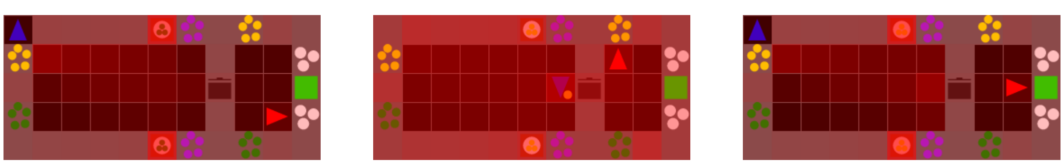
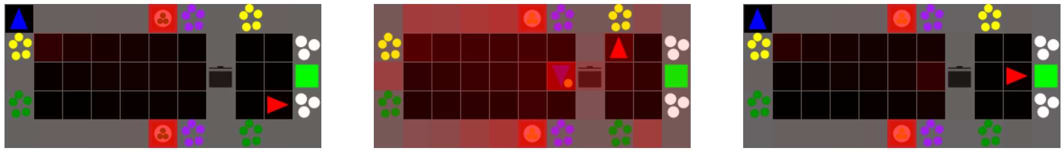

(a) (b) (c)

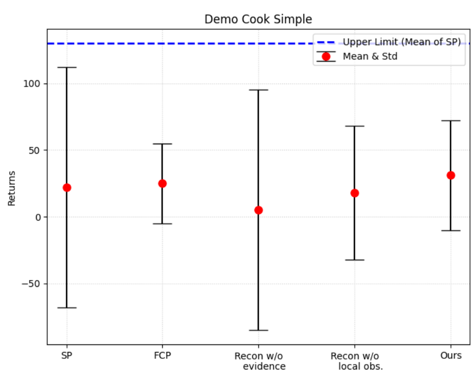
Figure 3: Reconstruction results with visualization of normalized epistemic uncertainty on Demo Cook Simple layout. (A) The initial frame reflects high epistemic uncertainty of recipe. (B) After few interactions uncertainty of recipe relatively decrease. (C) Resampling after successful delivery brings direct epistemic uncertainty.



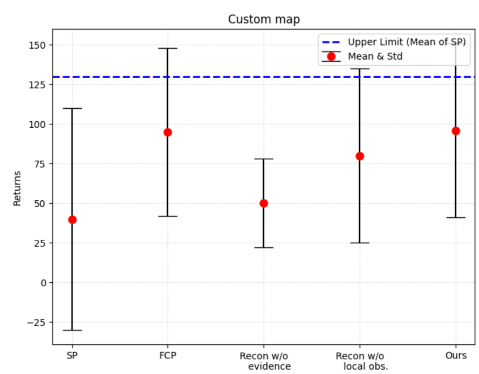
(a) (b) (c)

Figure 4: Reconstruction results with visualization of normalized epistemic uncertainty on Demo Cook Simple layout. (A) The initial frame reflects high aleatoric uncertainty of recipe. (B) After few interactions uncertainty of recipe slightly decrease. (C) Resampling after successful delivery brings direct aleatoric uncertainty.

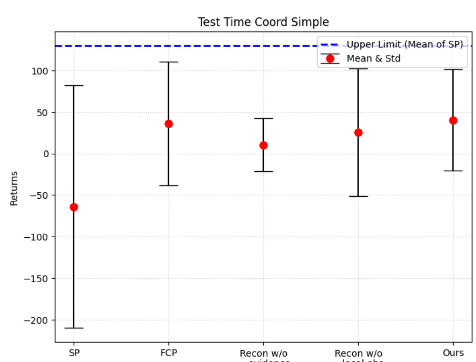
3 and 4 show changes of epistemic and aleatoric uncertainty during roll out. The both uncertainty is high when it has no evidence to predict in initial frame and frame after delivery is done. But, it decreases slightly after few interactions which collects evidence of its prediction. It reflects our assumption reconstruction with uncertainty is implicit way to model partner because agent predict only with seen or unseen behavior of partner while agent never see a recipe directly.



(a)



(b)



(c)

Figure 5: Cross-play results with unseen partners for each layout (A) Demo Cook Simple (B) Custom map (C) Test Time Coord Simple

Quantitative results of ZSC 5 shows that our method outperform slightly than FCP but, it is even worse when without local information and evidence because it heavily rely on wrong prediction. Although ours achieve best score among baselines, it can not tell we solve this problem since score gap between FCP is quite small and even invisible. Thus, the algorithm needs to be modified for further improvement.

5 Related work

Zero-shot and Ad-hoc Teamwork. Zero-Shot Coordination (ZSC) emphasizes population diversity (Strouse et al., 2021; Lupu et al., 2021; Zhao et al., 2023; Yu et al., 2023; Sarkar et al., 2023; Cui et al., 2021) to promote varied strategies for breaking symmetry (Hu et al., 2020; Lupu et al., 2021; Zhao et al., 2023; Sarkar et al., 2023). Cui et al. (2021) formalize this with k-level reasoning, while Yu et al. (2023) argue for incorporating human-like biases by training partners with randomly initialized linear reward functions, an implicit form of theory of mind. Addressing cooperative incompatibility, Li et al. (2023) proposes weighting harder-to-coordinate agents more heavily during sampling. Yan et al. (2023) achieves diversity using a single partner by injecting noise into actions. More recently, McKee et al. (2022) showed that environmental diversity often outweighs population diversity, and Jha et al. (2025) attained better performance with a single partner trained across diverse environments.

While Ad-hoc Teamwork (AHT) (Stone et al., 2010) shares fundamental motivations and characteristics with ZSC, literature within the AHT domain has predominantly focused on agent modeling (Albrecht & Stone, 2018). Standard approaches often involve inferring a partner's intent or strategy from a set of pre-defined "types" (Barrett & Stone, 2015; Albrecht & Stone, 2019) or adapting policies online based on interaction histories (Gu et al., 2022). Specifically, methods like PLASTIC (Barrett & Stone, 2015) and ODA (Gu et al., 2022) aim to select the best response from a library of known policies. However, these methods typically assume that the difficulty lies in identifying the partner's strategy rather than perceiving the environment itself. Our work distinguishes itself by addressing scenarios where the primary coordination bottleneck is partial observability of the environment (e.g., the recipe), requiring the agent to deduce the global state from the partner's behavior.

Uncertainty Estimation in Deep Learning. Reliable decision-making in physical environments requires agents to know what they do not know. Traditionally, Bayesian Neural Networks (BNNs) (MacKay, 1992; Neal, 1995) serve as the principled framework for this purpose by placing a prior distribution over model weights. Blundell et al. (2015) and Gal & Ghahramani (2016) introduced practical approximations such as Bayes by Backprop and Monte Carlo (MC) Dropout, respectively. However, these stochastic methods necessitate multiple forward passes (sampling) during inference to estimate the posterior, incurring a high computational cost that hinders real-time applicability in fast-paced multi-agent interactions. To address this latency issue, Deep Evidential Learning (DEL) (Sensoy et al., 2018; Amini et al., 2020) proposes a deterministic approach. Instead of sampling weights, DEL treats the network outputs as parameters of a higher-order distribution (e.g., Dirichlet distribution for classification), enabling the explicit quantification of both aleatoric (data noise) and epistemic (model ignorance) uncertainty in a single forward pass. While uncertainty has been utilized in RL primarily for exploration (Osband et al., 2016) or risk-sensitive control, we leverage DEL to prevent the agent from becoming overconfident in its state reconstruction under partial observability.

6 Further Improvement

Instead of conclusion, we write further improvement since our work is not concluded. In my perspective, the failure to advance remarkably is caused by following reasons.

1. **Environment is not well designed**: In contras to original Overcooked benchmark(Carroll et al., 2019), OvercookedV2(Gessler et al., 2025) focuses on communication protocol between agents. But communication channel is too restricted for training and evaluating these kind of coordination challenge. Thus it force to need heuristic for solving this work.
2. **Model is not optimized** : There are many of hyper-parameters since it combines many modules with pre-train, fine-tune regime. The visual results sometime shows the prediction is incomplete where the circumstance human can predict correctly. Thus, lack of ablation degrades its performance.
3. **Algorithmic incompleteness** : Instead of naively concatenating uncertainty, there will be potential way to adapt it to model such as masking when uncertainty is too high. Also, we can do better in classification task with multi-hot encoding while we used one-hot encoding for fixed category classification.

With respect to those reasons, we will do more ablation on model architecture and optimization and move to other environments where reconstruction is more valuable. If we test on the environment where looking around is cheap, we also can try to uncertainty-driven curiosity as we planned earlier. By the way, we will conduct more effort to improve this research

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